06b — Spelling (English)

What it detects: misspelled **ordinary English words** in a document's printed text — "Certifcate", "Univeristy", "Recieved". Genuine issuers print from proofread templates; a retyped or fabricated document often carries a typo. Small issuers make mistakes too, so this is a **weak signal**: every confirmed misspelling is shown (with its box on the page), but the rule that scores it is low-weight.

Code: `services/forensics/spelling.py` (`analyze_spelling`), run by `tasks/document_processing.py` (`_run_spelling_check` , right after font consistency — it needs that task's OCR words and extracted fields). Every document, PDF or image. One `document_checks` row: `spelling_consistency` . Dictionaries: `pyspellchecker` (word-frequency lists, bundled — no network).

English only. An Arabic pass was built and tried on real documents, but its results were not good enough, so Arabic words are skipped (`CHECKED_LANGUAGES = (ENGLISH,)` in `spelling.py` ; adding `ARABIC` turns it back on). Language is decided **per word**, so a mixed English / Arabic document still has its English words checked; the text-layer agreement test below is also computed from English words only.

How noise is kept out

Two passes. The dictionary pass is cheap and strict about what it lets through; the language model judges only what survives.

1. Dictionary pass — what becomes a candidate

An OCR word is checked only if it is:

Filter	Why
Latin letters only (Arabic and mixed-script words skipped); no digits or symbols (`@ # / . _ …`)	codes, IDs, emails, amounts
≥ 4 letters; not an acronym (all caps ≤ 5); not CamelCase	abbreviations, brands
not in an extracted field that is a name, address, ID, contact detail or code (`student_name` , `address` , `iban` , `trn` , …) or the reference number	names are not dictionary words
not handwriting (OCR styleFont), not ≥ 50 % inside a detected signature or stamp	signature ink, curved stamp text
OCR confidence ≥ 0.85	an uncertain read is more likely an OCR error

It becomes a **candidate** only if it is:

- **unknown in every accepted variant** — British spellings (`colour` , `organisation` , `programme` , `enrolment` …) and a small regional allow-list (`Sharjah` , `IGCSE` , `KHDA` …);
- **in the PDF's own text layer as read**, on a page whose text layer agrees with OCR on ≥ 60 % of its words (else the layer is not trusted). A word the PDF spells correctly but OCR misread is dropped here. Scans and images have no layer — only the OCR-confidence filter applies;
- **one edit away from a common word** (frequency ≥ 100). An unknown word with no near common word is a name or a foreign word, not a typo.

At most 30 distinct candidates (most frequent first) go on to the model.

2. Language-model judgment

One call per document (`LLMService.judge_spelling` , temperature 0), each candidate with ±6 words of context and its dictionary suggestions. Verdicts: `misspelling` , `valid_word` , `valid_variant` , `proper_noun` , `abbreviation_or_code` , `transliteration_or_foreign` , `ocr_misread` , `unsure` . The prompt tells it that a false alarm is worse than a miss.

Flagged: only `misspelling` at **medium or high** confidence, with a correction that differs from the word. Everything else is listed in the scope note (`data.set_aside`), not scored.

No model configured: nothing is flagged; result `limited` , the unconfirmed words noted. The dictionary alone is too noisy to score.

Result

```
{"result": "flag" | "pass" | "limited" | "not_applicable",
 "details": [
   {"finding": "spelling_error", "severity": "low", "page": 1,
    "bounding_box": {...}, "description": "Page 1: 'Certifcate' looks misspelled — 'Certificate' (English).
…",
    "data": {"text", "correction", "language": "en", "context", "model_confidence",
             "ocr_confidence", "confirmed_by_text_layer"}},
   …one per occurrence…,
   {"finding": "spelling_scope", "severity": "info",
    "data": {"words_checked", "words_checked_by_language", "unknown_words", "candidates",
             "text_layer_pages", "set_aside": [{"word", "verdict"}], "misspellings"}}]}
```

`not_applicable` : no printed English word was read (e.g. an Arabic-only document).

Scoring

Rule `content.spelling_error` (migration `a2d6f9c3e7b4`): any `spelling_error` finding → **weight 5, severity low**, fired **once per document** however many words (one typo is shown, but ten typos do not add up to a high score). Editable in Settings › Risk Rules like any rule.

Where it shows

- Case page: check "Spelling (English)" — headline "2 misspelled words: 'Certifcate', 'Univeristy'", one item per word with its correction; a **sky-blue box** around each occurrence on the page.
- Report: in the checks table, as a highlight region (sky blue) and as an exception line per word.